

AI System Design Copilot – Module-wise Blueprint

Module 1: Learn (Interactive System Design Explorer)

This module focuses on helping users understand how real-world distributed systems are designed. Instead of simply showing diagrams, the platform explains the reasoning behind every architectural decision.

Features

- **AI Architecture Generation** – Generate complete High-Level Design (HLD) architectures for popular systems such as WhatsApp, Netflix, Uber, Instagram, and Amazon.
- **Interactive Architecture Canvas** – Explore generated architectures where every component is clickable and connected through visual data flow.
- **Component Deep Dive** – Learn why each service or technology is used, what problem it solves, its advantages, disadvantages, alternatives, and real-world use cases.
- **Architecture Evolution Timeline** – Visualize how famous systems evolved over time, from monoliths to large-scale distributed architectures.
- **Technology Tradeoff Explorer** – Compare technologies such as PostgreSQL vs MongoDB or Redis vs Memcached based on scalability, consistency, latency, cost, and real-world usage.
- **Architecture Knowledge Base** – Access a built-in learning hub containing explanations of databases, caches, queues, load balancers, CDNs, messaging systems, and other distributed system concepts.
- **Case Studies** – Explore real-world engineering decisions taken by companies like Netflix, Meta, Uber, Amazon, and Google.

Module 2: Practice (Architecture Builder & Simulator)

This module allows users to design their own system architectures, validate them with AI, and understand how systems behave under different traffic conditions.

Features

- **Drag & Drop Architecture Builder** – Build custom HLD

diagrams using servers, databases, caches, queues, gateways, storage systems, and networking components.

- **AI Architecture Review** – Analyze user-created designs to detect scalability issues, missing components, Single Points of Failure (SPOFs), bottlenecks, and architectural improvements.
- **Architecture Scorecard** – Evaluate designs based on scalability, availability, reliability, maintainability, security, complexity, and cost efficiency.
- **Traffic Scenario Simulator** – Simulate different traffic situations such as Black Friday, IPL Final, Flash Sales, Product Launches, and Viral Events to observe system behavior.
- **Failure Simulation (Chaos Mode)** – Simulate failures by disabling components like Redis, Kafka, or databases and visualize how failures propagate throughout the architecture.
- **Optimization Suggestions** – Receive AI-powered recommendations such as adding caching, read replicas, sharding, message queues, CDNs, or autoscaling based on system requirements.
- **Performance Analysis** – Estimate bottlenecks, latency, throughput, and scalability limitations under different workloads.

Module 3: Design (Low-Level Design & UML Studio)

This module helps users practice object-oriented design, UML diagrams, and software design principles required for technical interviews and production systems.

Features

- **UML Workspace** – Design Class Diagrams, Sequence Diagrams, Use Case Diagrams, Activity Diagrams, State Diagrams, Component Diagrams, and ER Diagrams.
- **AI LLD Review** – Review UML diagrams using SOLID principles, OOP concepts, design quality, coupling, cohesion, and maintainability metrics.
- **Design Pattern Recommendations** – Suggest suitable design patterns such as Singleton, Factory, Strategy, Observer, Adapter, Builder, and Decorator based on the design.
- **Code Generation** – Generate starter code from approved UML diagrams in languages such as Java, C++, Python, and C#.

- **Reverse Engineering** – Generate UML diagrams from uploaded source code to understand existing projects.
- **Diagram Optimization** – Recommend cleaner class structures, improved relationships, and better object-oriented practices.

Module 4: Interview Mode (AI Interview Simulator)

This module provides a realistic system design interview environment where users solve problems under interview conditions and receive detailed feedback.

Features

- **AI Mock Interviewer** – Conduct complete HLD and LLD interviews with dynamic follow-up questions.
- **Requirement Gathering Simulation** – Evaluate how users collect functional and non-functional requirements before designing a system.
- **Live Architecture Evaluation** – Assess system designs while they are being built and challenge users with changing requirements.
- **Adaptive Follow-up Questions** – Introduce real interview scenarios such as sudden traffic spikes, service failures, regional expansion, and feature additions.
- **AI Feedback Report** – Generate detailed reports highlighting strengths, weaknesses, missing concepts, and suggested improvements.
- **Progress Tracking** – Track interview scores, completed problems, improvement areas, and learning progress over time.

Module 5: Community & Collaboration

This module enables collaborative learning, knowledge sharing, and peer feedback.

Features

- **Architecture Sharing** – Publish HLD and LLD designs for the community.
- **Peer Review System** – Allow users to review, comment on, and improve each other's architectures.
- **Community Templates** – Browse reusable system design templates for common applications.
- **Leaderboards & Challenges** – Participate in weekly design challenges and compare solutions with other engineers.
- **Version History** – Maintain multiple versions of architectures and

compare design improvements over time.

Vision

The goal of the platform is not to become another diagram-generation tool, but to serve as an **AI System Design Copilot** that helps engineers **learn, practice, simulate, review, and master High-Level Design (HLD) and Low-Level Design (LLD)** through interactive architecture building, AI-guided explanations, real-time simulations, and interview-oriented feedback.

Fonts

- Headings → Plus Jakarta Sans
- Body → Inter
- Code → JetBrains Mono
- LLD/HLD Canvas → Virgil

Color Pallet

- **#C2DED1** (Mint)
 - **#ECE5C7** (Warm Cream)
 - **#CDC2AE** (Soft Beige)
 - **#354259** (Navy)